PROPOSAL
RISET KOLABORASI INDONESIA
PRIORITAS RISET NASIONAL
SKEMA A

JUDUL PAYUNG PENELITIAN

Optimising Droplet Hydrodynamics and Heat Transfer for Next Generation Indonesian Data Centre

SUB – JUDUL PENELITIAN

Applying spray cooling to heat exchangers

Peran: Prof. Dr-Ing. Ir. Nasruddin,M.Eng. (UI)

Mitra 1

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Peneliti Mitra	: 1. Prof. Dr-Ing. Ir. Nasruddin, M. Eng. (UI)
	2. Prof. Tirto Prakoso, S.T., M. Eng., Ph.D. (ITB)

UNIVERSITAS INDONESIA
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DAFTAR ISI

1 RINGKASAN PROPOSAL

Indonesia’s digital transformation has sharply increased the demand for data centers, which are now essential for cloud services, AI, and big data. However, cooling systems remain a major challenge. Conventional air cooling is inefficient, and liquid cooling often requires high energy input, leading to high costs and carbon emissions. In Indonesia’s tropical climate, this makes data center operation even more demanding. This project proposes spray cooling as an innovative solution. By using droplets to transfer heat more efficiently, spray cooling offers lower energy use and improved reliability. The research will study droplet dynamics—such as size, speed, frequency, and spreading behaviour—under different working fluids and surface conditions. It will also explore how coatings, nanostructures, and surface designs influence cooling effectiveness. A key part of the work is the development of advanced droplet generators that can deliver controlled and repeatable droplets.

The project will be led by Universitas Gadjah Mada, with strong collaborations from Universitas Indonesia and Institut Teknologi Bandung. These partners will contribute expertise in energy systems, refrigeration, chemical engineering, and renewable fluids. The collaboration will also broaden the scope by evaluating spray cooling applications in other energy systems. Expected outcomes include scientific publications, prototype devices, and practical guidelines for spray cooling in data centers. The project also provides training opportunities for students and young researchers, ensuring knowledge transfer and capacity building. In the long term, this research supports Indonesia’s national goals for digital infrastructure and aligns with the UN Sustainable Development Goals on energy efficiency, innovation, and climate action.

2 PENDAHULUAN

2.1 Latar belakang masalah

In recent years, Indonesia’s rapid digital transformation has been accompanied by a sharp rise in the demand for reliable, secure, and energy-efficient data centers. The expansion of Industry 4.0, cloud services, big data analytics, and artificial intelligence (AI) has placed data centers at the very heart of the nation’s digital infrastructure. The importance of data centers is also highlighted in the National Long-Term Development Plan 2022–2045, which sets clear targets for their establishment. Their role is further reinforced by major national strategies such as *Satu Data Nasional*, which seeks to unify government data, and the National AI Roadmap, which positions advanced computational systems as a driver of innovation and competitiveness.

Despite their strategic importance, Indonesian data centers face distinctive challenges arising from high energy consumption and the tropical climate. Notably, the largest share of operational energy is consumed by the thermal management system (Huang et al., 2019), which is still dominated by conventional air-cooling methods. This approach, however, is less efficient (Niemann et al., 2013; Patankar, 2010).(Niemann et al., 2013; Patankar, 2010). In addition, liquid-cooling systems often require substantial power input and strongly relies on the liquid properties (Azarifar et al., 2024). This raises operating costs and increases carbon emissions, making data-center cooling a major challenge that needs to be addressed.

Advanced two-phase cooling has emerged as a promising alternative to conventional methods due to its superior heat dissipation and lower energy requirements (Karayiannis & Mahmoud, 2017; Mudawar, 2013; Sefiane & Koşar, 2022). Among them, spray cooling is a notable approach that uses droplets for efficient thermal management. These methods exploit the superior heat transfer properties of droplets and thin liquid films to achieve high cooling capacity (Cheng et al., 2016). However, practical implementation in the Indonesian context requires a deeper understanding of the fundamental fluid dynamics and thermal behaviour of droplets on engineered surfaces.

Although extensive research has been carried out over the past two decades, several knowledge gaps remain unresolved. Liang and Mudawar (Liang & Mudawar, 2017) provided a comprehensive review of droplet impact on hot surfaces as a fundamental event in spray cooling. Some of the main challenges include the lack of advanced diagnostics and models for droplet impact and evaporation as well as and the lack of agreement on critical heat flu. In addition, surface properties such as nano-coating, wettability, and inclination angle can drastically influence droplet cooling performance but have not been adequately mapped in a data center relevant setting. Finally, the control system of spray cooling, particularly to cover the intermittent cooling and varied environment, also become a challenge (Xu et al., 2022). Those problem shows that the comprehensive investigation on this topic are urgently needed.

The aim of this research is to build a fundamental understanding of droplet hydrodynamics and heat transfer as a basis for developing efficient and reliable cooling technologies for next-generation Indonesian data centers. The study will investigate droplet behaviour under various conditions and working fluids, assess the influence of surface properties on cooling performance, and explore different droplet generation methods. In the last decade, the principal investigator's research group have been actively engaged with the topic of droplet impacting on hot surface (Deendarlianto et al., 2014, 2016, 2018; Wibowo et al., 2021). Equally important, the research group also investigate the heat transfer topic related to the cooling system (Kusumaningsih et al., 2025; Pranoto et al., 2023; Widyatama et al., 2023, 2024). This experience and preliminary knowledge form a strong foundation to further explore the implementation of spray cooling systems for data centers. Furthermore, the proposed research will be strengthened through collaboration with partners from Universitas Indonesia and Institut Teknologi Bandung. This collaboration will help strengthen the study by providing complementary expertise and expanding the scope of the study as well as the impact.

In terms of United Nations Sustainable Development Goals (SDGs), this research is aligned with SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 13 (Climate Action), as it aims to improve energy efficiency, strengthen digital infrastructure, and reduce the environmental impact of data center cooling. More importantly, the topic also lies at the intersection of two of the six research priority areas of the EQUITY grant: energy and digitalisation, which also support the national research priority areas. As discussed previously, reliable data centers are a critical backbone of digitalisation. In addition, cooling systems account for the largest share of energy use and are expected to further increase overall consumption in the future. These facts highlight the importance of understanding this topic to support the development of efficient data center cooling systems, particularly in Indonesia.

2.2 Tujuan

The long term goal of this research is to support the development of efficient and reliable cooling technologies for next-generation Indonesian data centers. In general, the objectives of this study are focused as follows

1. To study droplet hydrodynamics and heat transfer under different sizes, frequencies, and velocities. The study also considers the effect of various working fluids, including binary mixtures.
2. To examine how surface properties influence droplet behaviour and cooling. Materials, coatings, and nanostructures will be tested to optimise wettability, spreading, and heat transfer.
3. To explore different droplet generation methods for controlled droplet production. Vibration is considered one of the most promising options, but other approaches will also be evaluated.

3 METODOLOGI

The research will be mainly conducted at the Fluid Mechanics Laboratory, Department of Mechanical and Industrial Engineering, Universitas Gadjah Mada. Over the past decade, the proposing research group has focused extensively on droplet studies, supported by experimental facilities relevant to this topic, as illustrated in Figure 1. In general, the set-up consists of the main rig, heating system, specimen, droplet generator, and data measurement devices.

The main rig is used to place the specimen, droplet generator, and data measurement devices together. Due to the importance of the surface inclination on the droplet behaviour, this rig has been designed to enable easy adjustment of the inclination as needed. Next, the droplet generator is equipped with eight different nozzles (Figure 1 b) to allow both single and multiple droplets to be investigated. It is also supported with pressurised air system to control the droplet impact characteristics.

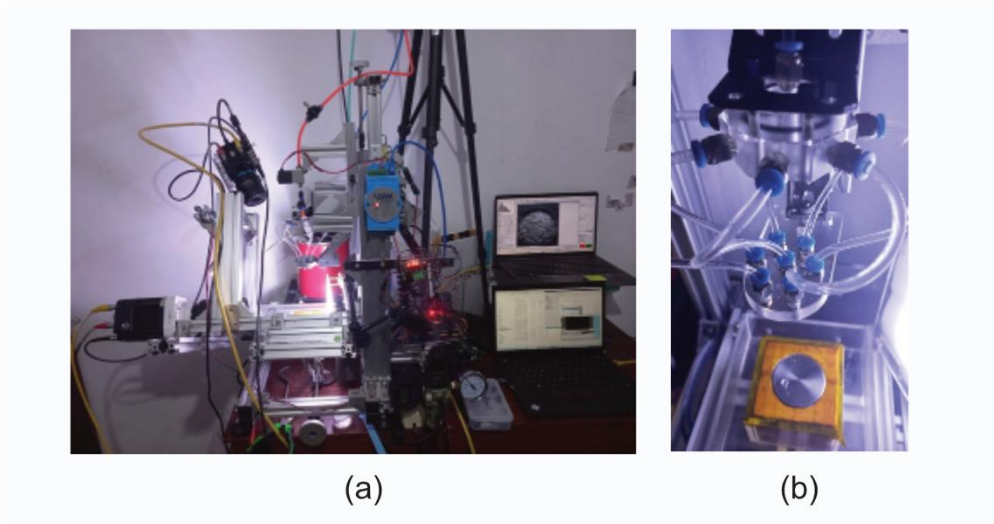

Figure 1. Current experimental devices

The heating system is installed separately to provide flexibility for the main rig. It consists of heater elements and a holder for the solid specimen. The experimental setup is also equipped with various measurement devices. Two high-speed video cameras are used to record the droplet dynamics simultaneously. The first camera, a Phantom Miro 200, is positioned parallel to the solid surface to capture contact behaviour and measure droplet parameters. The second camera is placed at an upward angle of about 60° to provide a wider field of view.

4 RENCANA PENELITIAN

Figure *2* shows the research framework and the collaborative scheme in the proposed research. This study focuses on three important areas: advancing spray cooling for next generation data centers in Indonesia, developing environmentally friendly working fluids, and exploring the broader implementation of spray cooling for sustainable energy systems.

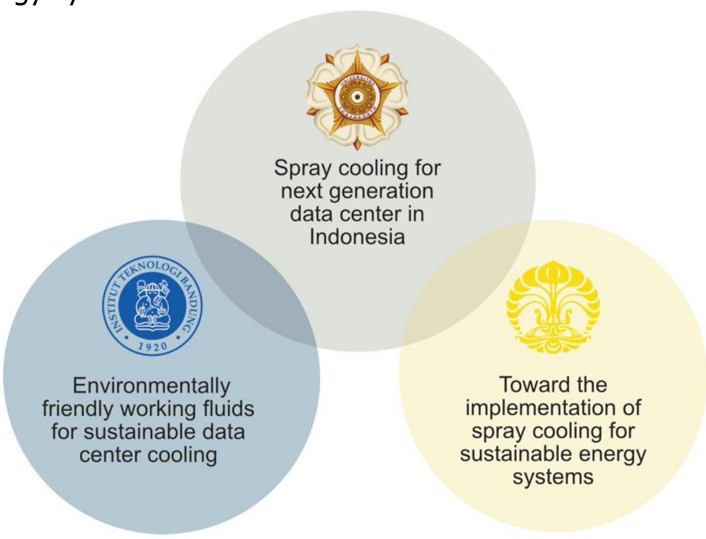

Figure 2. Research scope and the collaboration scheme

(a) Pelaksanaan penelitian di PT-Pengusul

The proposed research aims to investigate the droplet behaviour impacting the hot surface for data center applications. The main activities are listed as follows.

A. Upgrading of the Current Facility of the Droplet Impacting the Hot Surface

The initial step of this study is to upgrade the existing droplet research facility. Over the past decade, continuous improvements have been made to enhance data quality and deepen the analysis. In this research, several steps will be conducted. First, the rig and the liquid path system will be upgraded. This allows a more surface configuration to be used. In addition, one of our future objectives is to investigate the behaviour of multiple droplets. Hence, another crucial aspect is to develop a new droplet generator that can produce multiple droplets uniformly.

B. Experimental Study of Droplet on Various Operational Conditions

The experimental study of droplet behaviour is one of the main work packages in this project. This part of the research is directed at exploring how droplets interact with heated surfaces under a variety of conditions. By conducting systematic experiments, the study will highlight the key features of droplet dynamics that are often difficult to capture in theoretical or numerical approaches. Careful observation will ensure that each stage of the droplet interaction is well documented.

The main objective is to produce high quality data that represents droplet phenomena across a wide range of operational conditions. Parameters such as droplet size, impact velocity, and surface temperature will be adjusted one by one to see their influence. In addition, combinations of these parameters will be explored to reveal more complex behaviour. Through this approach, the research will be able to map the conditions that strongly affect spreading, recoiling, or evaporation of droplets.

The experimental data will then be analysed to identify patterns and key trends in droplet behaviour. This includes comparing how different operational conditions influence spreading, recoiling, and evaporation. The analysis will also look for critical thresholds, for example the point where a droplet fully evaporates or starts to bounce. By combining these observations, the study can provide a clearer explanation of the mechanisms that govern droplet interaction.

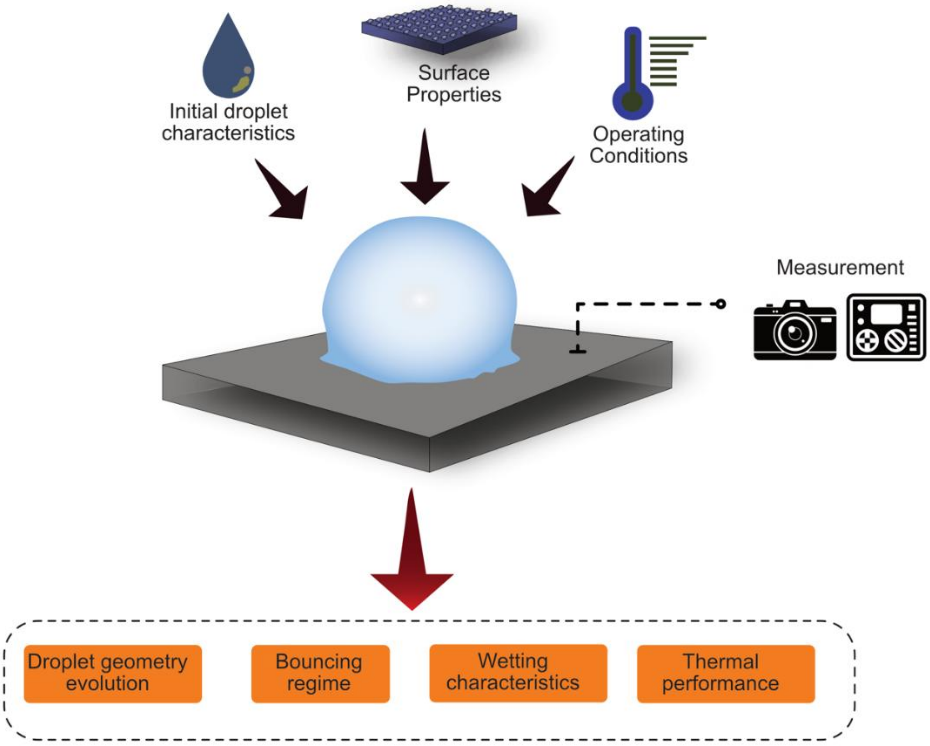

Figure 3. Experiment on droplet behaviour scheme

The outcomes of this study are expected to provide useful insights into the fundamental mechanisms of droplet interaction with solid surfaces. The collected data will serve as a reliable reference not only for this project but also for future studies in related fields. In the longer term, these findings can contribute to improving heat transfer models and guiding the design of applications where droplet cooling plays an important role.

C. Examining the Surface Modification Method to Improve the Heat Transfer

Surface modification is an important method to enhance heat transfer. By altering the surface structure or adding a special coating, the interaction between the liquid and the solid can be controlled. This directly influences how droplets spread, evaporate, or rebound during contact, which in turn affects the overall cooling performance.

The study will compare different types of surface modifications, such as changes in roughness, wettability, or the use of nano-coatings. Each modified surface will be tested under controlled conditions to observe its effect on droplet behaviour and thermal response. These comparisons will make it possible to identify the surfaces that provide the most significant improvement in heat transfer.

The results will then be analysed to establish clear links between surface properties and heat transfer enhancement. Attention will be given to determining which surface features are most effective, whether hydrophilic, hydrophobic, or hybrid. The findings are expected to give valuable insights that can support the design of more efficient cooling technologies in the future.

D. Early development of the Droplet Generator

The development of a droplet generator is an essential step in this project. The generator is expected to produce droplets with controlled size, frequency, and velocity. Having a stable and repeatable droplet source is important to ensure that the experiments can be carried out under consistent conditions. This step also helps to reduce uncertainties in the later analysis of droplet behaviour. It should be noted that previous studies (Cheng et al., 2016; Liang & Mudawar, 2017) have provided several methods that can be used to generate the droplet, which can be used as a starting point of this work package.

In the early stage, the focus will be on designing a simple but effective system. One of the most promising methods is using nozzle vibration, which can create uniform droplets when combined with a controlled liquid supply. The syringe pump will be used to adjust the flow rate, while the vibration of the nozzle helps control both droplet frequency and velocity. Other options, such as pressure-driven injection or piezoelectric actuation, may also be explored if needed.

The outcome of this development will provide a strong foundation for further improvement. Once a stable single-droplet system is achieved, the design can be extended to produce multiple droplets or to operate under more complex conditions. In the long run, the droplet generator will become a key tool to support systematic studies of droplet dynamics and heat transfer.

(b) Pelaksanaan penelitian di PT-Mitra (maksimum 1 halaman tiap peneliti mitra)

1) Universitas Indonesia

To enhance the research knowledge and impact of the proposed study, several collaborations will be established The first collaboration is with Prof. Dr.-Ing. Ir. Nasruddin, M.Eng., a Professor at the Department of Mechanical Engineering, Universitas Indonesia. His expertise covers energy systems optimisation, refrigeration engineering, and thermodynamics, with a strong track record in developing efficient and sustainable energy conversion technologies. He has also been actively involved in research on CO_2 adsorption, renewable-based cooling systems, and thermoeconomic analysis of energy systems. With this background, Prof. Nasruddin will contribute valuable knowledge and experience to support the development of advanced and energy-efficient cooling strategies.

This collaboration will focus on evaluating the possibility of applying spray cooling to other energy systems, such as heat exchangers, residential and building cooling, or solar power plants. Based on his expertise in energy systems optimisation and refrigeration engineering, Prof. Nasruddin will lead the technical evaluation to explore how spray cooling can be integrated into these applications. The study will not only consider the feasibility of implementation but also assess potential benefits in terms of energy efficiency, reliability, and operational cost reduction. Such an evaluation is important to understand the wider opportunities of spray cooling beyond its use in data centers.

In the next stage, several potential materials will be carefully selected to represent the typical components found in these systems. Their interaction with spray cooling will be tested to provide practical insights into material compatibility and thermal performance. This step is expected to provide valuable data that can guide the design of future prototypes. Ultimately, the collaboration aims to broaden the application of spray cooling technology across multiple energy systems and contribute to the development of more sustainable energy solutions in Indonesia.

2) Institut Teknologi Bandung

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6 INDIKATOR KEBERHASILAN (TARGET CAPAIAN)

NO	INDIKATOR KEBERHASILAN	JUMLAH	DESKRIPSI
1	Keluaran (*output*) Hasil Riset	1–2 international journal publications (Scopus Q1/Q2),	Scientific publications in reputable journals and international conferences, along with the development of a prototype/experimental device to support future research.
2	Dampak (*outcome*) Hasil Riset	1 early guidance related to spray cooling data center	Tangible impact of the research through the development of an initial guideline or reference document on spray cooling technology for data center applications and further research
3	Pembinaan *peer*	2 students supervised / involvement of early career researchers	At least one of the early researchers, who has just obtained the doctoral degree will gets involved. Furthermore, at least two students will also join this project.
4	*Networking* internasional		

7 JADWAL PELAKSANAAN

Table 1. Timeline of proposed research

Task ID	TITLE	2025			2026						
		Okt	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
a	Literature review	■									
b	Experimental Set Up	■	■	■							
c	Set up validation and running test		■	■							
d	Surface preparation		■	■	■	■					
e	Surface characterisation			■	■	■	■				
f	Experiment on droplet impacting hot surface			■	■	■	■	■	■		
g	Droplet generation method design	■	■	■							
h	Manufacturing process			■	■	■	■				
i	Initial trial						■	■	■		
j	Design improvement								■	■	■
h	Scientific article and report writing						■			■	■

8 PETA JALAN (*ROAD MAP*) RISET

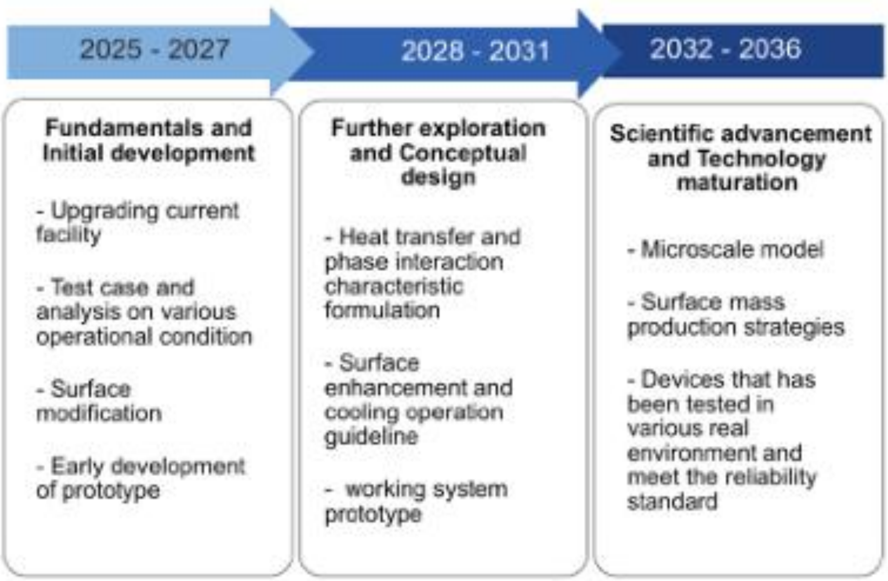

Figure 4. Ten years roadmap of the droplet for data center cooling

This figure illustrates a ten-year roadmap outlining the phased development of droplet-based cooling technologies for data centers. The first phase (2025–2027) focuses on fundamentals and initial development, including upgrading the current facility, testing cases under various operational conditions, surface modification, and early prototype development. The second phase (2028–2031) moves into further exploration and conceptual design, emphasising detailed studies on heat transfer and phase interaction, guidelines for surface enhancement and cooling operations, and working system prototypes. Finally, the third phase (2032–2036) highlights scientific advancement and technology maturation, aiming to establish microscale models, mass-production strategies for advanced surfaces, and devices tested in real environments that meet reliability standards. This roadmap positions droplet cooling as a scalable and sustainable solution aligned with long-term technological goals.

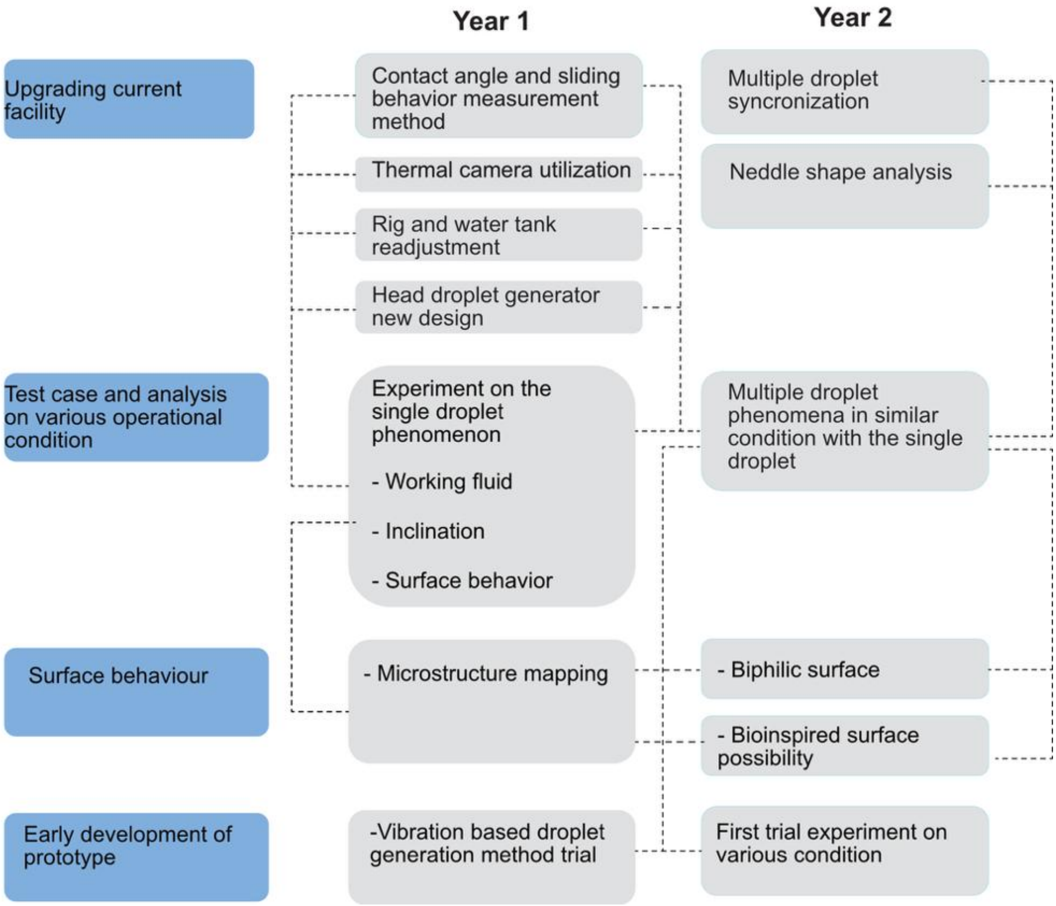

Figure 5. Research plan for the current study

This figure presents a two-year research roadmap tailored to the proposed study, breaking down specific tasks into measurable milestones. In Year 1, the focus is on upgrading the current facility by integrating a contact angle and sliding behaviour measurement method, thermal camera utilisation, re-adjustment of the rig and water tank, and redesigning the droplet generator head. Concurrently, experiments on single-droplet phenomena (working fluid, inclination, surface behaviour) and microstructure mapping are conducted, supported by trials of a vibration-based droplet generation method. In Year 2, the scope expands to multiple-droplet synchronisation, needle shape analysis, and comparison of multi-droplet phenomena with single-droplet behaviour. Further, advanced surfaces such as biphilic and bioinspired designs are explored, alongside first-trial experiments under diverse conditions. This structured plan ensures systematic progression from fundamental single-droplet analysis to more complex multi-droplet scenarios, forming a strong foundation for future large-scale implementation.

9 USULAN BIAYA

Table 3 shows the funding allocation for research over a one year period, with a total proposed amount of Rp 200.000.000,00.

Table 2. Funding Allocation for Research

No	Item	Unit	Number	Unit price	Total cost
Bahan dan peralatan					
1	Profil Aluminium & Akrilik	Set	1	2.000.000	2.000.000
2	Pipa Tembaga & Sirip Aluminium	Set	2	4.000.000	8.000.000
3	Baut, mur, dan fitting	Set	1	500.000	500.000
4	Spray Nozzle Array Head (custom)	Unit	4	5.000.000	20.000.000
5	Vibration Actuator & Signal Generator	Set	1	28.000.000	28.000.000
6	Pompa Roda Gigi (Gear Pump)	Unit	1	9.000.000	9.000.000
7	Syringe Pump System	Unit	1	8.000.000	8.000.000
8	Sistem Reservoir & Pipa	Set	1	4.000.000	4.000.000
9	Data Acquisition System (DAQ)	Unit	1	20.000.000	20.000.000
10	Termokopel Tipe-K (set)	Set	2	3.000.000	6.000.000
11	Mass Flow Meter (Coriolis)	Unit	1	15.000.000	15.000.000
12	Pressure Transducer	Unit	2	2.500.000	5.000.000
13	Power Supply DC	Unit	1	14.000.000	14.000.000
14	Fluida Kerja (Air Deionisasi, Etanol)	Paket	1	3.000.000	3.000.000
15	Kabel dan komponen elektronik	Paket	1	10.000.000	10.000.000
Mobilisasi					
1	Jakarta – Yogyakarta	Paket	2	4.500.000	9.000.000
2	Jakarta – Bandung	Paket	2	4.500.000	9.000.000
Lainnya					
1	Honorarium asisten	Bulan	10	1.200.000	12.000.000
2	Jasa Fabrikasi & Modifikasi Permukaan	Paket	1	10.000.000	10.000.000
3	Biaya Publikasi Jurnal Internasional (APC)	Artikel	1	5.000.000	5.000.000
4	ATK dan Keperluan Riset	Paket	1	2.500.000	2.500.000
Total keseluruhan					**Rp 200.000.000**

10 CV PENELITI

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1. Nama pengusul : Prof. Dr. Ir. Deendarlianto, S.T., M. Eng.

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RISET KOLABORASI INDONESIA
PRIORITAS RISET NASIONAL

Saya yang bertanda tangan di bawah ini, menyatakan bahwa:

Nama Peneliti Mitra	: Prof. Dr-Ing. Ir. Nasruddin, M. Eng.
NIP	: 197204111995121001
Universitas/Institut	: Universitas Indonesia
Fakultas / Sekolah	: Fakultas Teknik
Program Studi	: Teknik Mesin

Dengan ini menyatakan bahwa saya **bersedia** bermitra dalam Program Riset Kolaborasi Indonesia (RKI) Prioritas Riset Nasional (PRN) 2025 – 2026 dengan:

Nama Peneliti Pengusul	: Prof. Dr. Ir. Deendarlianto, S.T., M. Eng.
NIP	: 197208032008121001
Universitas / Institut	: Universitas Gadjah Mada
Fakultas/Sekolah	: Fakultas Teknik (Departemen Teknik Mesin dan Industri)
Judul Penelitian	: Optimising Droplet Hydrodynamics and Heat Transfer for Next Generation Indonesian Data Centre

Demikian surat kesediaan ini dibuat dengan sebenarnya tanpa ada paksaan dari pihak manapun.

Mengetahui,
Direktur Pendanaan dan Ekosistem Riset

UNIVERSITAS INDONESIA
PUSAT ADMINISTRASI

Prof. Apt. Rani Sauriasari, M.Med.Sc., Ph.D
NUP. 031003026

Depok, 29 – 9 - 2025
Yang menyatakan,

Prof. Dr-Ing. Ir. Nasruddin, M. Eng.
NIP. 197204111995121001

Rencana Anggaran Biaya (RAB)

Judul	: Optimising Droplet Hydrodynamics and Heat Transfer for Next Generation Indonesian Data Centre
Pengusul	: Prof. Dr. Ir. Deendarlianto, S.T., M. Eng.
Skema	: Skema A
Fakultas	: Fakultas Teknik

Rekapitulasi RAB

Deskripsi	Biaya (Rp)
Honor (maksimal 25%)	12.000.000
Biaya Bahan Habis Pakai	152.500.000
Biaya Perjalanan	18.000.000
Biaya Operasional Lainnya	17.500.000
Total	200.000.000

Rincian RAB8

Deskripsi	Justifikasi Pemakaian	Kuantitas	Harga Satuan (Rp)	Biaya (Rp)
Honor				
Honorarium Asisten Peneliti (Mahasiswa S2/S3)	Membantu persiapan eksperimen, pengambilan data, dan analisis awal.	10 bulan	1.200.000	12.000.000
			SUBTOTAL (Rp)	12.000.000
Biaya Bahan Habis Pakai				
A. Pembuatan Test Section Heat Exchanger				
Profil Aluminium & Akrilik	Konstruksi rig utama dan chamber pengujian	1 Set	2.000.000	2.000.000
Pipa Tembaga & Sirip Aluminium	Fabrikasi beberapa prototipe heat exchanger (compact fin-tube).	2 Set	4.000.000	8.000.000
Baut, mur, dan fitting	Perakitan rig dan sambungan fluida.	1 Set	500.000	500.000
B. Sistem Spray Cooling				
Spray Nozzle Array Head (custom)	Kepala generator droplet yang didesain untuk cakupan area heat exchanger.	4 Unit	5.000.000	20.000.000

Deskripsi	Justifikasi Pemakaian	Kuantitas	Harga Satuan (Rp)	Biaya (Rp)
Vibration Actuator & Signal Generator	Mengontrol frekuensi dan karakteristik droplet (pulsating spray).	1 Set	28.000.000	28.000.000
Pompa Roda Gigi (Gear Pump)	Memompa fluida kerja secara kontinyu ke sistem spray.	1 Unit	9.000.000	9.000.000
Syringe Pump System	Kalibrasi dan pengujian presisi dengan laju alir rendah.	1 Unit	8.000.000	8.000.000
Sistem Reservoir & Pipa	Penampungan dan sirkulasi fluida kerja.	1 Set	4.000.000	4.000.000
C. Instrumentasi dan Akuisisi Data				
Data Acquisition System (DAQ)	Merekam data dari berbagai sensor secara simultan.	1 Unit	20.000.000	20.000.000
Termokopel Tipe-K (set)	Mengukur temperatur permukaan heat exchanger dan fluida.	2 Set	3.000.000	6.000.000
Mass Flow Meter (Coriolis)	Mengukur laju alir massa fluida kerja (spray) dengan akurasi tinggi.	1 Unit	15.000.000	15.000.000
Pressure Transducer	Mengukur tekanan pada saluran fluida.	2 Unit	2.500.000	5.000.000
Power Supply DC	Catu daya untuk pemanas (heater) pada heat exchanger.	1 Unit	14.000.000	14.000.000
D. Bahan Habis Pakai				
Fluida Kerja (Air Deionisasi, Etanol)	Cairan yang akan diuji untuk spray cooling.	1 Paket	3.000.000	3.000.000
Kabel dan komponen elektronik	Wiring sensor dan aktuator ke sistem DAQ dan kontrol.	1 Paket	10.000.000	10.000.000
			SUBTOTAL (Rp)	152.500.000
Biaya Perjalanan				
Perjalanan Jakarta	Koordinasi dan	2 Paket	4.500.000	9.000.000

Deskripsi	Justifikasi Pemakaian	Kuantitas	Harga Satuan (Rp)	Biaya (Rp)
- Yogyakarta (PP)	analisis bersama tim peneliti mitra di UGM.			
Perjalanan Jakarta - Bandung (PP)	Koordinasi dan analisis bersama tim peneliti mitra di ITB.	2 Paket	4.500.000	9.000.000
			SUBTOTAL (Rp)	**18.000.000**
Biaya Operasional Lainnya				
Jasa Fabrikasi & Modifikasi Permukaan	Pembuatan komponen custom dan pelapisan permukaan heat exchanger.	1 Paket	10.000.000	10.000.000
Biaya Publikasi Jurnal Internasional (APC)	Untuk publikasi hasil riset di jurnal bereputasi (Q1/Q2).	1 Artikel	5.000.000	5.000.000
ATK dan Keperluan Riset	Kebutuhan alat tulis dan material pendukung selama penelitian.	1 Paket	2.500.000	2.500.000
			SUBTOTAL (Rp)	**17.500.000**
Total Anggaran (Rp)				**200.000.000**

Menyetujui,
Wakil Dekan I Fakultas

Tanda tangan dan Cap

Prof. Dr. Ir. Yanuar, M.Eng., M.Sc.
NIP. 196001121987031003

Depok, 29 – 9 – 2025
Periset Utama

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